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Waiting for participants to join…



DATA PRESERVATION & SHARING

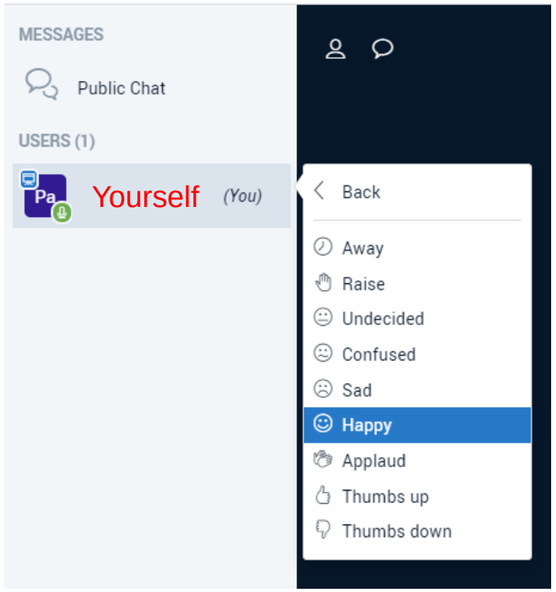
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Slides: https://osf.io/kdypa/

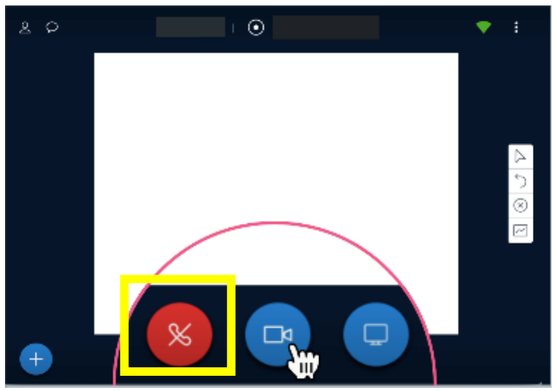
IS EVERYTHING CORRECT?

Can you hear the presenter and moderator?

a. If yes, set your status to ☺



b. If no, click on the telephone icon. It should not be blue



c. If you still have problems, set your status to 👎

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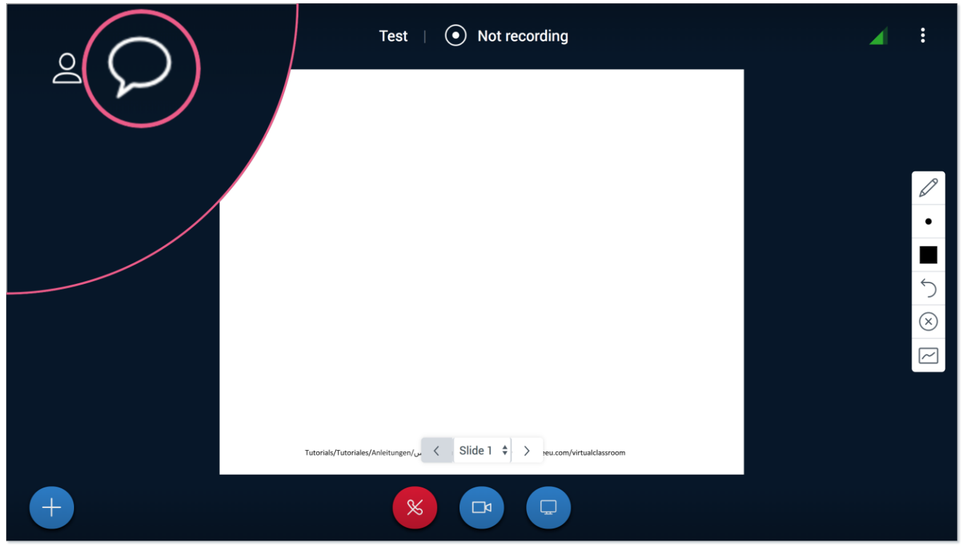
QUICK TROUBLESHOOTING TIPS

1. Reload the page
2. Restart the browser.
   - Do not "Restore all Tabs" when reopening the browser.
3. Open the meeting in a different browser.
4. Open the meeting in an Incognito or Private window.
5. Disconnect from any connected VPNs/wireless hotspots.
6. Leave meeting and re-join.
7. Restart your device.
8. Check email received before the session for more tips.

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HOW THE SESSION WORKS

1. 45’ presentation + 15’ questions
2. Write your questions in the public chat.
   - The moderator will collect them for the end of the session.
3. Participate in polls
4. Session is being recorded



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RESEARCH DEPARTMENT
UNIVERSITY LIBRARY

DATA PRESERVATION & SHARING

RDM webinar series, 16-06-2020

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Slides: https://osf.io/kdypa/

CC BY

OUTLINE

1. Preserving data
2. Sharing data

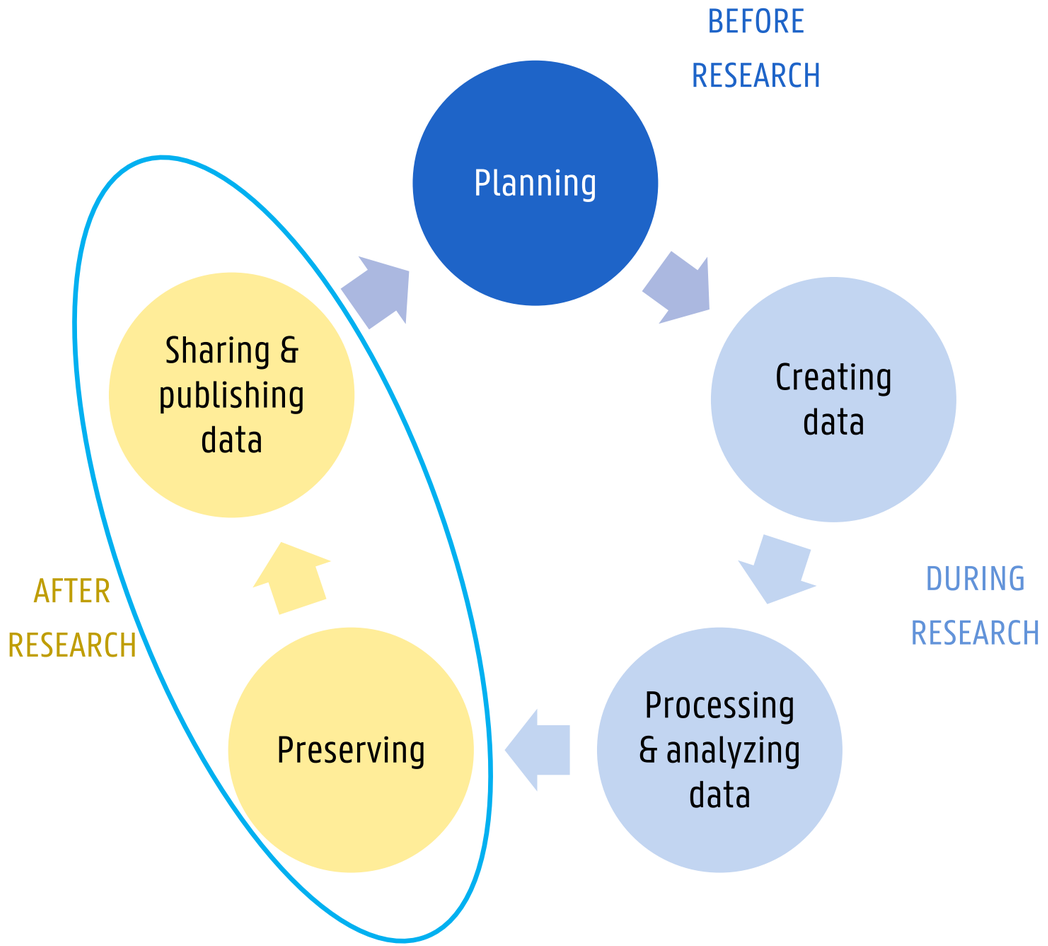
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PRESERVING DATA

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WHAT?

Preserving research data refers to the practice of keeping data available and usable in the longer term, beyond the end of your research project.



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PRESERVING != STORING & BACKING UP

Preserving

- milestone version
- ‘frozen’
- not in active use.

Requires actions to prevent data from becoming unavailable and unusable:

- Outdated software or hardware
- Storage media degradation
- A lack of sufficient documentation and structure

More than simply not deleting the data files created and stored during your research project!

WHY?

The main reasons for data preservation are:

- Ensuring that your research can be verified and reproduced
- Maintaining data for future reuse, e.g. for further research or teaching



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WHY?

Increasingly, funders, publishers, and institutions (including Ghent University) are requiring (certain) research data to be retained for a specified period and/or for a specific purpose.

Ghent University's RDM policy framework expects 'relevant' research data to be preserved for a minimum of 5 years.

ALLEA: “…secure preservation for a reasonable period”

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WHAT TO KEEP?

Selecting which (parts of) data to keep, and for how long, is an essential component of data preservation.

You have a key role in deciding what to retain and what not, as you know your data best.

Decisions may depend on factors such as:

- The type of data involved
- The norms in your discipline
- Purpose: potential future reuse, verification, or for other purposes

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WHAT TO KEEP – SELECTION CRITERIA

Common high-level criteria to guide decisions on what to keep:

- Legal/ethical requirements to keep data for a specified retention period (e.g. for clinical trials)
- Funder, institutional or publisher policies
- High potential reuse value of the data
- Great scientific, historical, or cultural significance of the data
- The data are unique and/or cannot easily be re-created.
- The benefits outweigh the costs of data preservation.

Note: there can also be valid reasons for not keeping (parts of) data (e.g. duplicate copies, intermediate steps, superseded versions, contractual obligations,…)

It is also important to document and justify your choices to keep or remove data, for example in your Data Management Plan.

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WHERE TO PRESERVE DATA?

Research data and documentation selected for retention should be kept in a suitable location and in a secure manner to ensure that they remain available and usable beyond the end of your project, with appropriate access rights.

Data repository

- Preferred

In house

- Sometimes it is not be possible/appropriate to deposit data in an external repository, e.g. for legal, ethical, contractual, practical, or other reasons.

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	Office 365			Shares		HPC	
Name	OneDrive FB	SharePoint	Teams SharePoint	H:/	Project Share (acl & bulk)	HOME/DATA/SCRATCH	VO_DATA/VO_SCRATCH
Location	Cloud	UGent	Cloud	UGent	UGent	UGent	UGent
Size	1TB	5GB	See link	5GB	GB-TB range	25 GB	GB-TB range
Backup mechanism	Versioning	Versioning	Versioning	snapshots	snapshots	none	none
Sync to local	✓		✓				
Confidential	Encrypt	✓	Encrypt	✓	✓	✓	✓
Intended use	Personal	Sharing	Sharing	Personal	Sharing	Personal	Sharing
Sharing external users?	✓	✓	✓				✓
Suitable for TB datasets					✓		✓
Data in use	✓	✓	✓	✓	✓	✓	✓
Data at rest					✓		

PREPARING DATA FOR PRESERVATION

Without associated information, research data quickly become useless. For all data selected for preservation, you should therefore keep a ‘data package’ consisting of:

- Research data files themselves
- Meaningful file/folder structure
- Documentation and metadata to ensure data remains findable, comprehensible, and (re)usable



Jørgen Stamp - Digitalbevaring.dk

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PREPARING DATA FOR PRESERVATION

Keeping data safe for the future therefore requires some preparation and effort on your part.

- Making available of sufficient documentation and metadata
- Converting data to file formats suitable for sustainable access
- Properly structure data files & folders
- Having the necessary rights and/or permissions in place to preserve and share data

Established domain-specific repositories will usually only accept data that meet their standards for file formats, documentation and metadata, data quality… Check in advance what the requirements are, so you can adequately prepare your data for deposit.

NON-DIGITAL RESEARCH DATA AND MATERIALS

- Consider whether digitizing the data is an option
- If not, your faculty, department, research group, lab etc. may offer local facilities to retain your data for verification or legal compliance purposes for a finite retention period.
- There are also repositories for non-digital materials you can make use of. E.g.:
  - BCCM (Belgian Co-ordinated Collections of Micro-Organisms): for the deposit of biological materials
  - Bioresource Center Ghent: offers support for biobanking projects as well as a central biobanking facility at Ghent University Hospital.

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> Data sharing results in science that is more accurate, more open, more knowledgeable, and has more resources at its disposal
>
> – Hendrick, 1988

SHARING DATA

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WHAT IS DATA SHARING?

- Sharing data from completed (parts of) research outside your project or research team.
- ≠ exchanging data with collaborators while your research is active.



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POLL

Are you open to sharing your research data?

1) Yes
2) No

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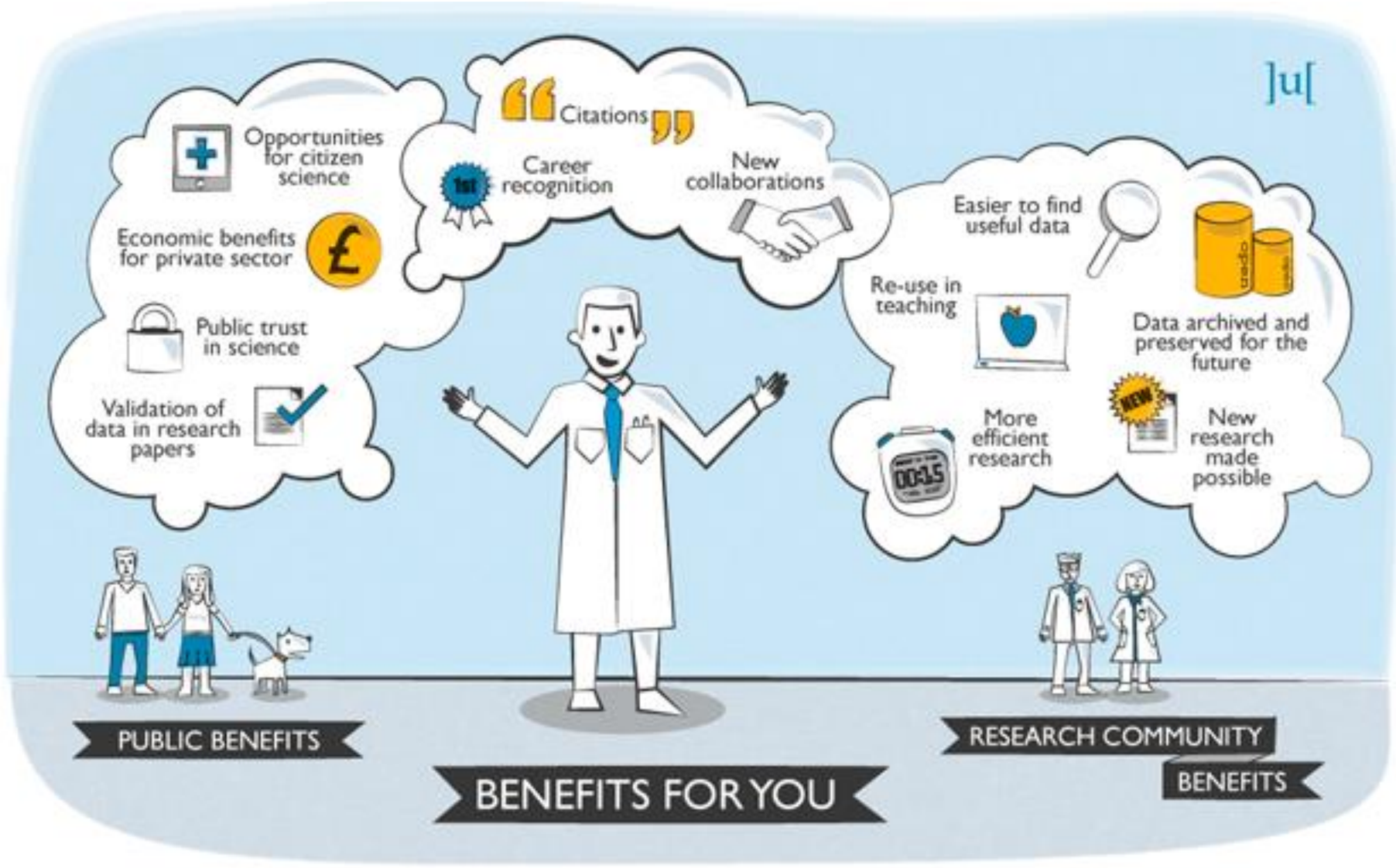
POLL

Are you open to (re)using shared data?

1) Yes
2) No

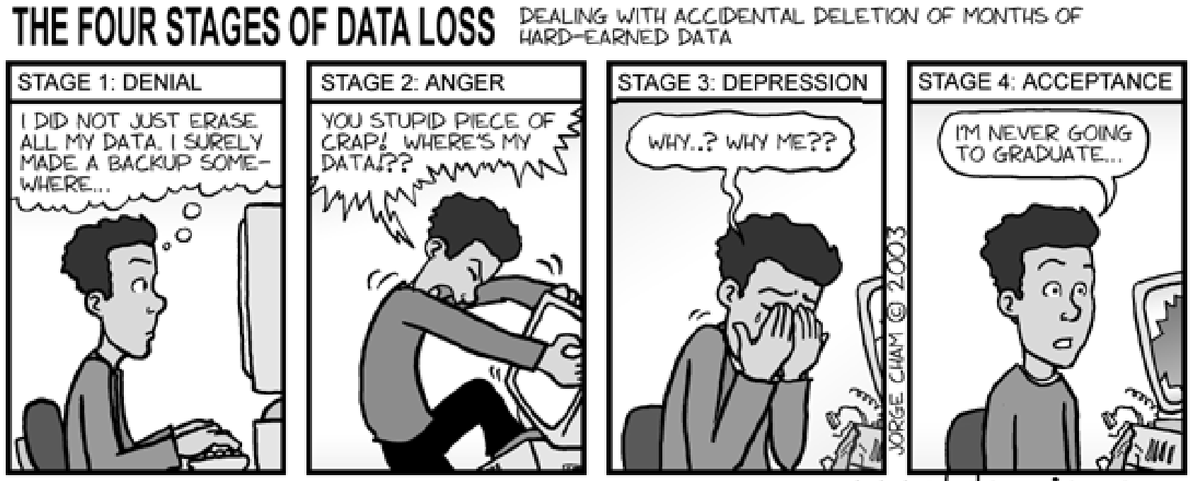
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WHY? BENEFITS OF MANAGING & SHARING DATA



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From 'Overcoming obstacles to sharing research data' by Brian Hole, licensed under CC-BY 4.0



"Piled Higeher and Deeper" by Jorge Cham, 'http://www.phdcomics.com/comics/archive.php?comicid=382

MINIMISE RISK OF DATA LOSS

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Most Scientific Research Data From the 1990s Is Lost Forever

A new study has found that as much as 80 percent of the raw scientific data collected by researchers in the early 1990s is gone forever, mostly because no one knows where to find it.

DANIELLE WIENER-BRONNER DECEMBER 23, 2013



https://www.theatlantic.com/national/archive/2013/12/scientific-data-lost-forever/356422/

How to share data for collaboration

Shannon E. Ellis
Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health
and
Jeffrey T. Leek *
Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health

September 1, 2017

Abstract

Within the statistics community, a number of guiding principles for sharing data have emerged; however, these principles are not always made clear to collaborators generating the data. To bridge this divide, we have established a set of guidelines for sharing data. In these, we highlight the need to provide raw data to the statistician, the importance of consistent formatting, and the necessity of including all essential experimental information and pre-processing steps carried out to the statistician. With these guidelines we hope to avoid errors and delays in data analysis.

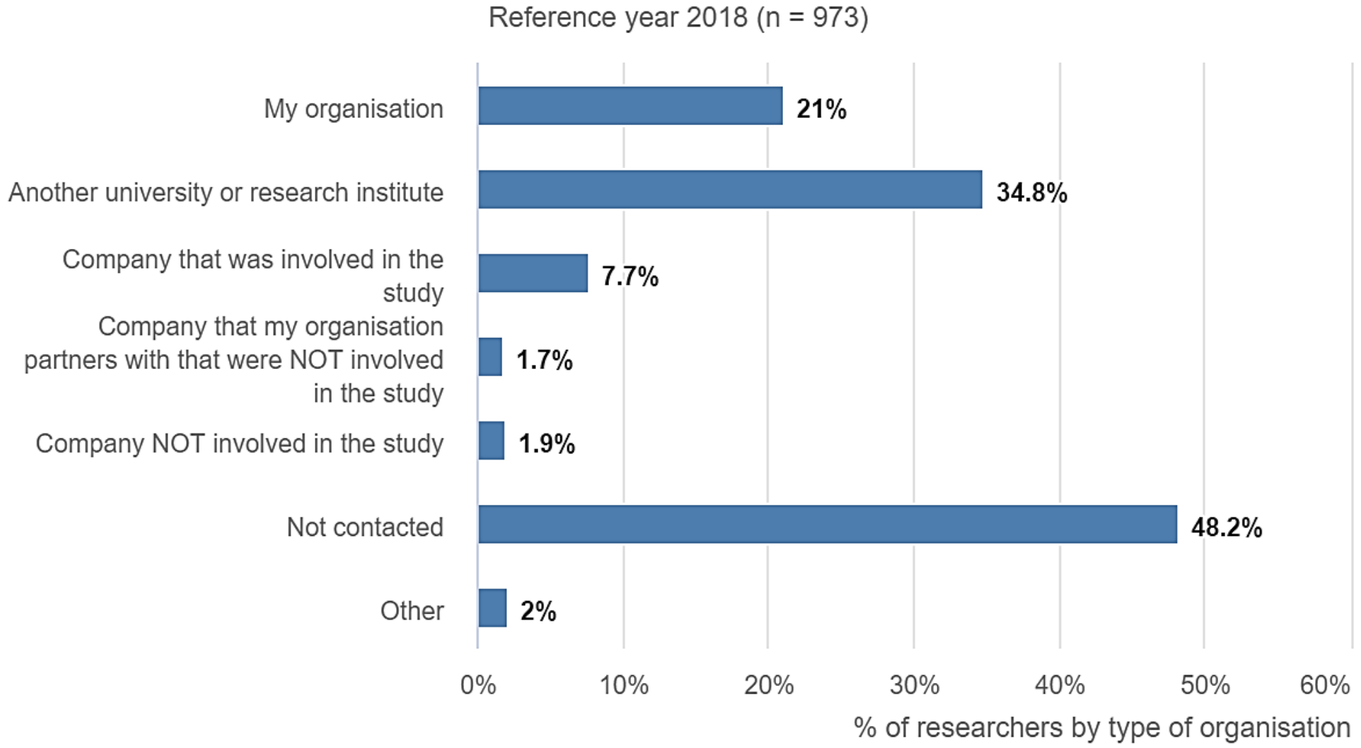
Keywords: tidy data, guidelines, statistician, data sharing, analysis

S. E. Ellis & J.T. Leek (2017), *PeerJ Preprints*, DOI: 10.7287/peerj.preprints.3139v5

FACILITATE COLLABORATION

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Consequences of data sharing: Contact made with researchers from outside their research team after sharing data, % of researchers by type of organisation

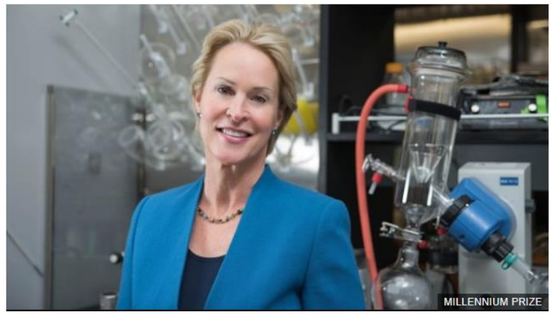


https://ec.europa.eu/info/research-and-innovation/strategy/goals-research-and-innovation-policy/open-science/open-science-monitor/facts-and-figures-open-research-data_en

Nobel Prize-winning scientist Frances Arnold retracts paper

3 January 2020 Share

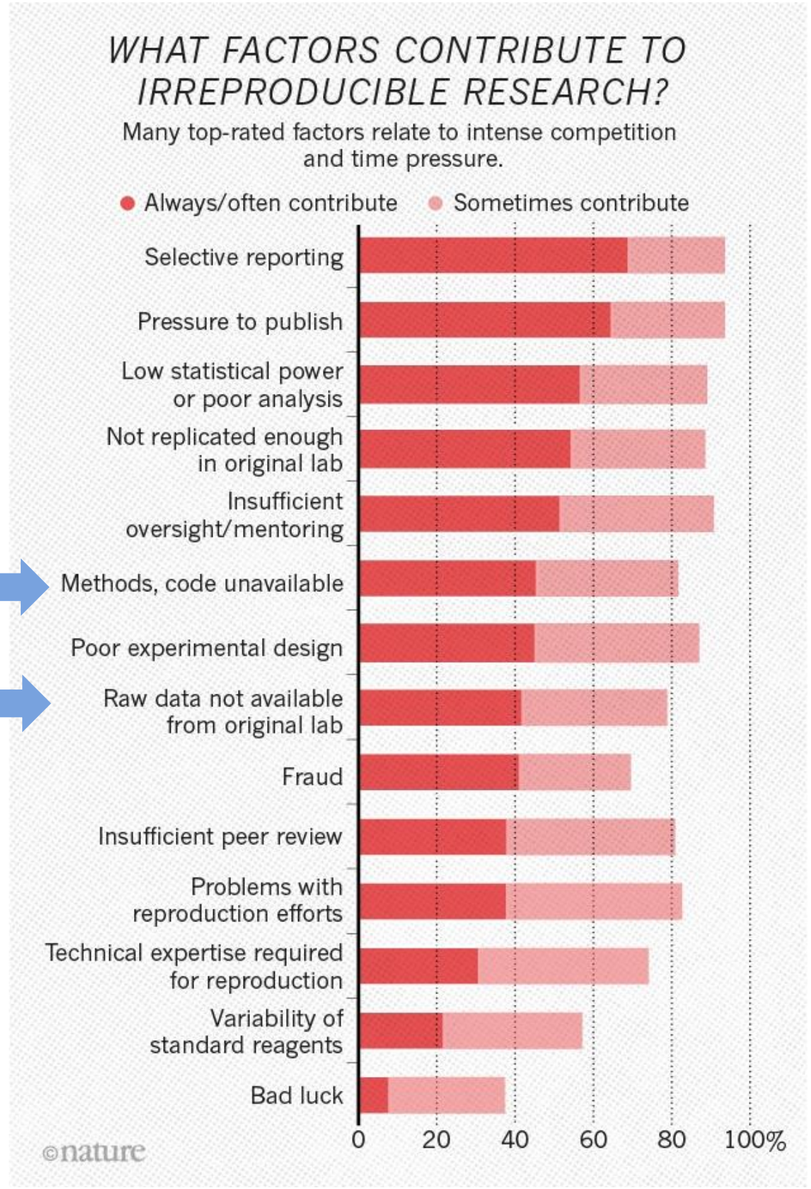
Nobel Prize



"It has been retracted because the results were not reproducible, and the authors found data missing from a lab notebook."

https://www.bbc.com/news/world-us-canada-50989423

SUPPORT RESEARCH INTEGRITY & REPRODUCIBILITY



M. Baker (2016), *Nature* 533, 452-454. DOI: 10.1038/533452a

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arXiv.org > cs > arXiv:1907.02565

Search...

Help | Advanc

Computer Science > Digital Libraries

The citation advantage of linking publications to research data

Giovanni Colavizza, Iain Hrynaszkiewicz, Isla Staden, Kirstie Whitaker, Barbara McGillivray

(Submitted on 4 Jul 2019 (v1), last revised 3 Oct 2019 (this version, v2))

Efforts to make research results open and reproducible are increasingly reflected by journal policies encouraging or mandating authors to provide data availability statements. As a consequence of this, there has been a strong uptake of data availability statements in recent literature. Nevertheless, it is still unclear what proportion of these statements actually contain well-formed links to data, for example via a URL or permanent identifier, and if there is an added value in providing such links. We consider 531,889 journal articles published by PLOS and BMC, develop an automatic system for labelling their data availability statements according to four categories based on their content and the type of data availability they display, and finally analyze the citation advantage of different statement categories via regression. We find that, following mandated publisher policies, data availability statements have become common by now, yet statements containing a link to a repository are still just a fraction of the total. We also find that articles with these statements, in particular, can have up to 25.36\% higher citation impact on average: an encouraging result for all publishers and authors who make the effort of sharing their data. All our data and code are made available in order to reproduce and extend our results.

G. Colavizza et al. (2019), arXiv: 1907.02565v2

"up to 25,36% higher citation impact on average" for articles linking to research data

ENHANCE RESEARCH VISIBILITY & IMPACT

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Building a Culture of Data Citation



'Building a culture of data citation' by ANDS, licensed under CC-BY 4.0

Science

Scientists are unraveling the Chinese coronavirus with unprecedented speed and openness



Medical workers of the Union Hospital with the Tongji Medical College of the Huazhong University of Science and Technology in Wuhan participate in a ceremony to form an "assault team" to battle a coronavirus epidemic in China. (Cheng Min/Xinhua/AP)

By **Carolyn Y. Johnson**

Jan. 24, 2020 at 10:17 p.m. GMT+1

Just 10 days after a pneumonia-like illness was first reported among people who visited a seafood market in Wuhan, China, scientists released the genetic sequence of the coronavirus that sickened them. That precious bit of data, freely available to any researcher who wanted to study it, unleashed a massive collaborative effort to understand the mysterious new pathogen that has been rapidly spreading in China and beyond.

https://www.washingtonpost.com/science/2020/01/24/scientists-are-unraveling-chinese-coronavirus-with-unprecedented-speed-openness/

ACCELERATE SCIENTIFIC DISCOVERY

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nature
International journal of science

Access provided by University Library Gent

Altmetric: 52 Citations: 55 More detail »

Editorial

Data's shameful neglect

Research cannot flourish if data are not preserved and made accessible. All concerned must act accordingly.

More and more often these days, a research project's success is measured not just by the publications it produces, but also by the data it makes available to the wider community. Pioneering archives such as GenBank have demonstrated just how powerful such legacy data sets can be for generating new discoveries — especially when data are combined from many laboratories and analysed in ways that the original researchers could not have anticipated.

(2009) *Nature* 461, 145. DOI: 10.1038/461145a

PLOS ONE

PUBLISH ABOUT BROWSE SEARCH

advanced search

Introduction
Minimal Data Set Definition
Acceptable Data Sharing Methods
Acceptable Data Access Restrictions
Unacceptable Data Access Restrictions
FAQs

Data Availability

Introduction

PLOS journals require authors to make all data necessary to replicate their study's findings publicly available without restriction at the time of publication. When specific legal or ethical restrictions prohibit public sharing of a data set, authors must indicate how others may obtain access to the data.

When submitting a manuscript, authors must provide a Data Availability Statement describing compliance with PLOS' data policy. If the article is accepted for publication, the Data Availability Statement will be published as part of the article.

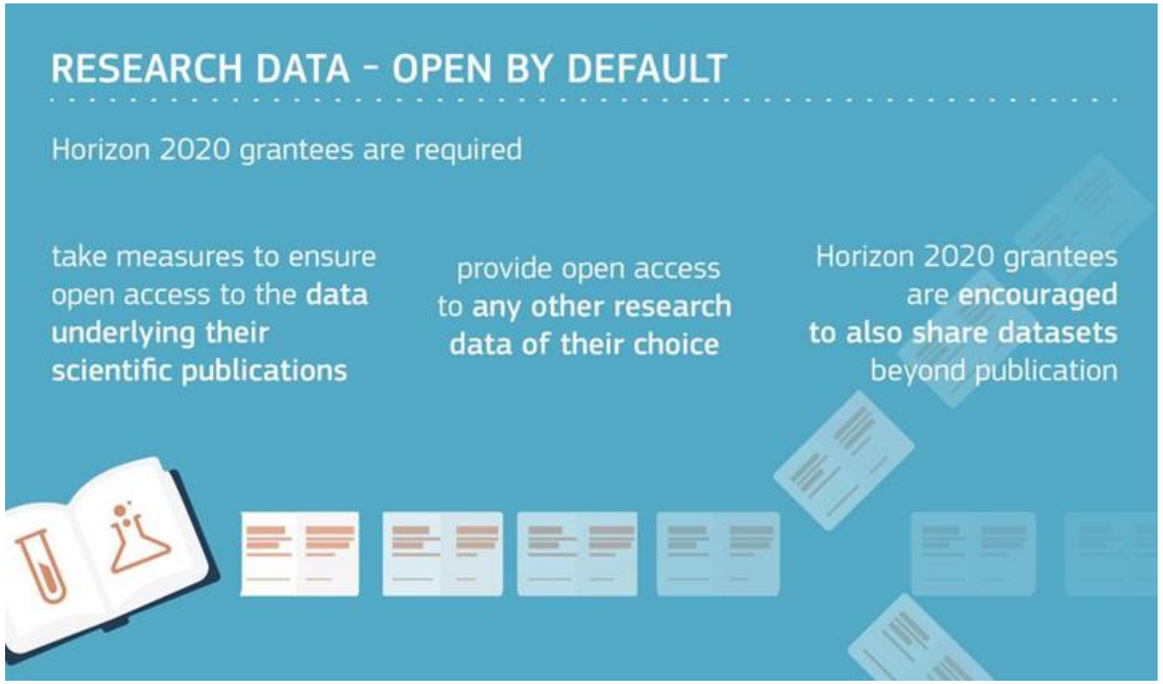
https://journals.plos.org/plosone/s/data-availability



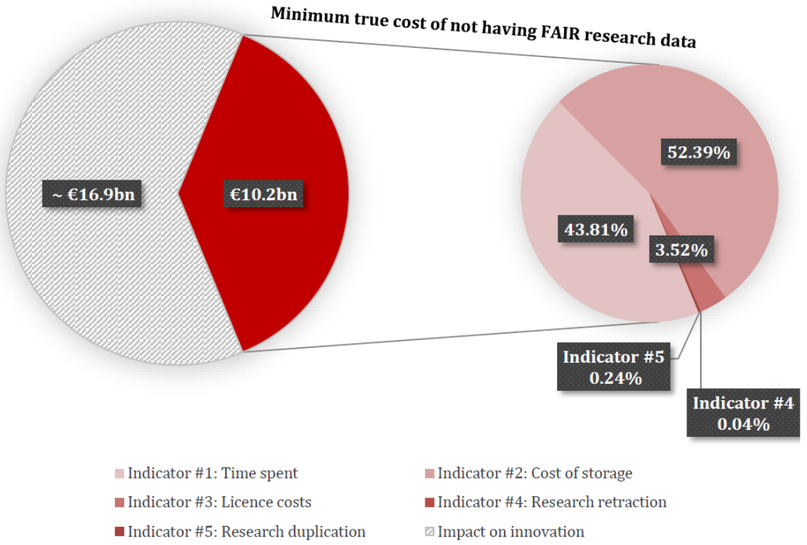
"gdpr" by medithIT, licensed under CC-BY 2.0

COMPLY WITH DATA REQUIREMENTS

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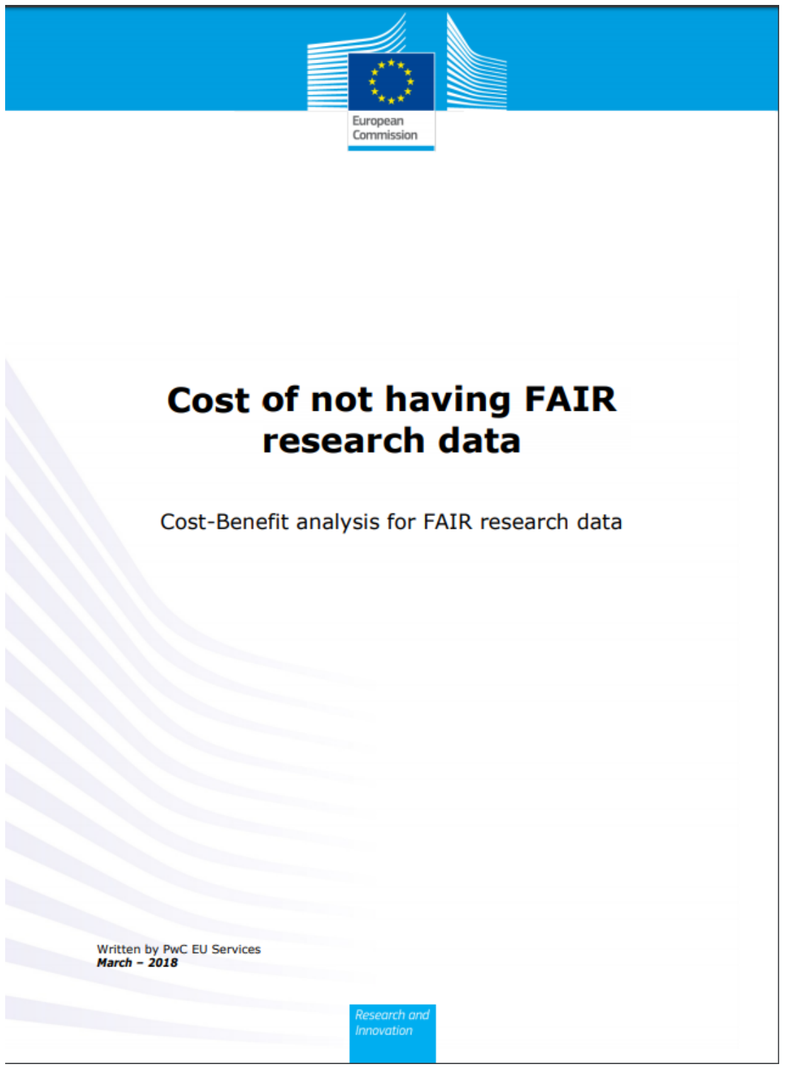


http://ec.europa.eu/research/images/infographics/policy/open-data-2016-w920.png



EFFICIENT USE OF RESOURCES

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"The cost of not having FAIR research data costs the European economy at least €10.2bn every year."

https://op.europa.eu/en/publication-detail/-/publication/d375368c-1a0a-11e9-8d04-01aa75ed71a1/language-en



Simon Hodson
@simonhodson99

Data from publicly funded research are a public good, i.e. non-rival and non-excludable - powerful statement of principle #idcc13 #jiscmrd

3:58 PM · Jan 16, 2013 · TweetDeck

https://twitter.com/simonhodson99/status/291559946494033920

PUBLICLY FUNDED RESEARCH = PUBLIC GOOD

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DATA SHARING

"Data Sharing" by Auke Herrema, licensed under CC-BY 4.0

CHALLENGES OF DATA SHARING

- Time and effort

  Making data sharable is labor intensive
- Lack of incentives for sharing data

  The current academic culture lacks the incentive structure to properly reward researchers who share their data.
- Loss of control over data

  Scoop, errors in their data, contradictory conclusions, misuse
- Personal data

  Anonymizing/pseudonymizing not always easy. Trade-off with utility.

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RESTRICTIONS ON DATA SHARING

Possible **reasons** for restricting the sharing of data are:

- Privacy
- Otherwise confidential or sensitive data
- Third-party data
- Data protected by copyright
- Data commercial/economic valorization potential

Legit reasons to opt out of open data requirements journals/funders

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DEGREES OF DATA SHARING

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

http://opendefinition.org

Limits on who can access & use data, how, or for what purpose

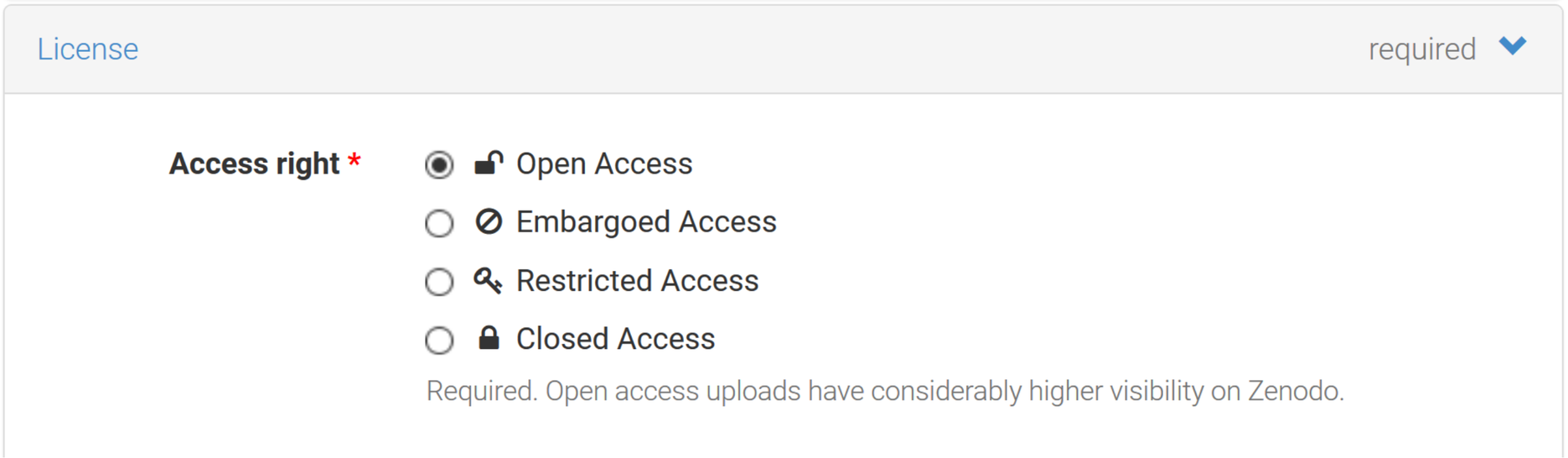
- only certain (types of) users
- only certain types of use
- …

Under embargo
Unable to share

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Adapted from 'Managing and sharing research data' by S. Jones, CC-BY

DEGREES OF DATA SHARING @ REPOSITORY



Access right settings at Zenodo repository

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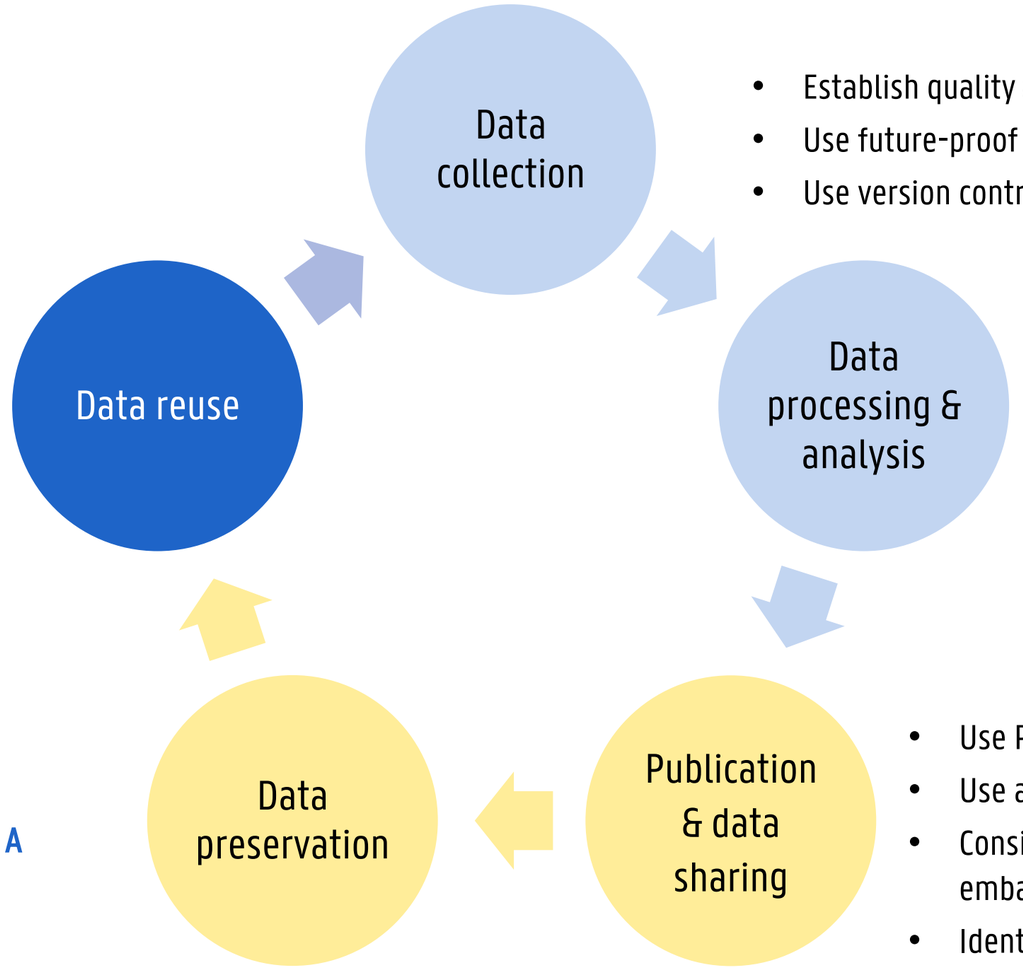
FAIR IN PRACTICE

F: Findable

A: Accessible

I: Interoperable

R: Reusable



- Establish quality assurance procedures. **I, R**
- Use future-proof file formats, follow name conventions. **I, R**
- Use version control. **R**

- Describe data & processes consistently. **R**
- Use (meta)data standards and controlled vocabularies for your discipline. **F, I**
- (Identify discipline-specific repositories*. **F, I, R**)

- Use PIDs to publish your data / research output. **A**
- Use author PIDs to distinguish yourself / your group. **F**
- Consider, choose and indicate machine-readable license, embargo periods. **A, I, R**
- Identify discipline-specific repositories*. **F, I, R**

Choose a trustworthy repository*. **A**

Cite data using data citation guidelines and persistent identifiers (PIDs, e.g. DOIs), **F**

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* Search for repositories at: FAIRsharing.org or re3data.org

WAYS OF DATA SHARING

- Mail data upon request (Avoid)
- Make them available via a personal or project website
- Add them as supplementary materials to a journal article
- Share data via a data repository/data archive (Preferred)



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PREFERRED WAY OF DATA SHARING

Share data via a trusted, (domain-specific) data repository

Advantages:

- No handling of data reuse queries and managing data access.
- Guarantees in terms of sustainable access to data.
- Make your data discoverable and citable.
- Make your data FAIR

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HOW TO SELECT A DATA REPOSITORY?

- Does your publisher recommend a repository?
- Does your domain have an established repository?
- Other
  - Certificates / ensure long-term preservation
  - Licenses / access rights management
  - Provides DOIs / persistent identifiers
  - Metadata standards
  - Costs
  - Criteria for selection of trustworthy repo's



Icons representing attributes of data repositories in Re3data.org, 10.1371/journal.pone.0078080

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CRITERIA FOR THE SELECTION OF TRUSTWORTHY REPOSITORIES

1. **Provision of Persistent and Unique Identifiers (PIDs)**
   a) Allow data discovery and identification
   b) Enable searching, citing, and retrieval of data
   c) Provide support for data versioning
2. **Metadata**
   a) Enable finding of data
   b) Enable referencing to related relevant information, such as other data and publications
   c) Provide information that is publicly available and maintained, even for non-published, protected, retracted, or deleted data
   d) Use metadata standards that are broadly accepted (by the scientific community)
   e) Ensure that metadata are machine-retrievable
3. **Data access and usage licences**
   a) Enable access to data under well-specified conditions
   b) Ensure data authenticity and integrity
   c) Enable retrieval of data
   d) Provide information about licensing and permissions (in ideally machine-readable form)
   e) Ensure confidentiality and respect rights of data subjects and creators
4. **Preservation**
   a) Ensure persistence of metadata and data
   b) Be transparent about mission, scope, preservation policies, and plans (including governance, financial sustainability, retention period, and continuity plan)

Source: https://www.scienceeurope.org/media/jezkhnoo/se_rdm_practical_guide_final.pdf

HOW TO FIND A DATA REPOSITORY?

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

seadatanet | Search

http://www.re3data.org

zenodo

B2SHARE EUDAT

DRYAD

Global Biodiversity Information Facility
Free and Open Access to Biodiversity Data

ads ARCHAEOLOGY DATA SERVICE

DataONE
Data Observation Network for Earth

Edinburgh DataShare

DataSTORRE
DataSTORRE: Stirling Online Repository for Research Data.

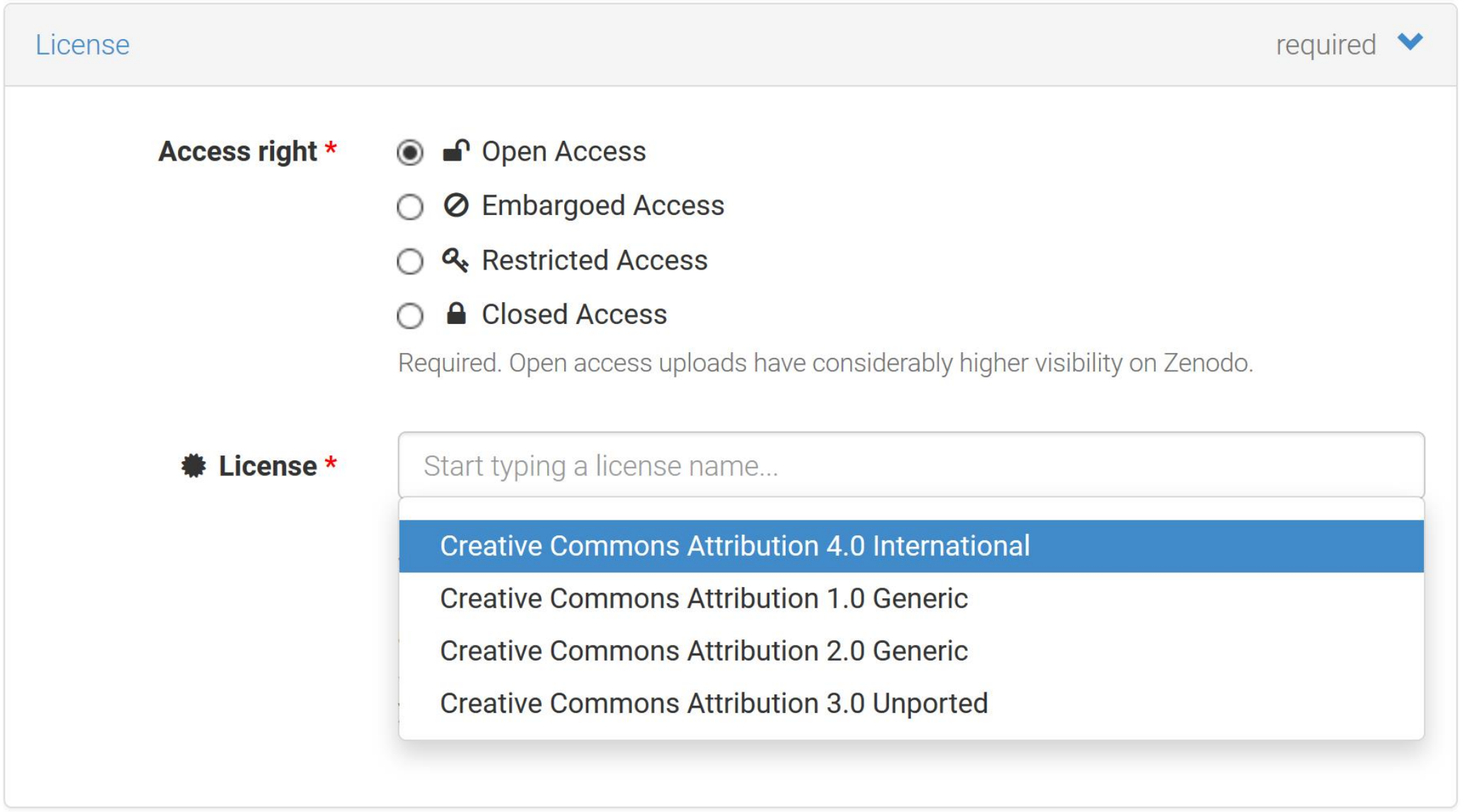
General purpose | **Domain-specific** | **Institutional**

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Watch a Re3data demo: https://www.fosteropenscience.eu/content/re3data-demo

LICENSING

- Let potential users know in advance what they can do with shared data.
- Trusted data repository: select license when depositing data.



Zenodo license chooser

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TAILOR-MADE LICENCES

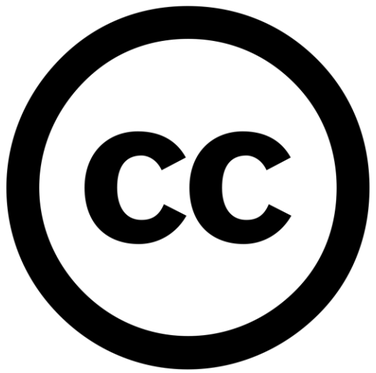
- Sometimes requirement from research partner, funder, research institution, data repository, etc.
- Most common reason to adopt them is commercial value of data.
- Written by the publisher or adapted from a standard license by adding / adaption conditions.
- Can increase complexity for users, decrease interoperability.
- Contact TechTransfer

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STANDARD LICENSES

- Terms and conditions are already defined.
- Might not cover very specific cases but multiple advantages:
  - Simplicity and transparency.
  - Machine readability and interoperability
- Examples:
  - Creative Commons (CC) and Open Data Commons (ODC).
  - Open Government licenses. E.g. UK Open Government license.
  - Open software licenses (e.g. GNU General Public License, Apache).





LICENSING: CREATIVE COMMONS CONDITIONS



Attribution. You let others copy, distribute, display, and perform your copyrighted work – and derivative works based upon it – but only if they give credit the way you request.



Non-Commercial. You let others copy, distribute, display, and perform your work – and derivative works based upon it – but for noncommercial purposes only



No Derivative Works. You let others copy, distribute, display, and perform only verbatim copies of your work, not derivative works based upon it.



Share Alike. You allow others to distribute derivative works only under a license identical to the license that governs your work.

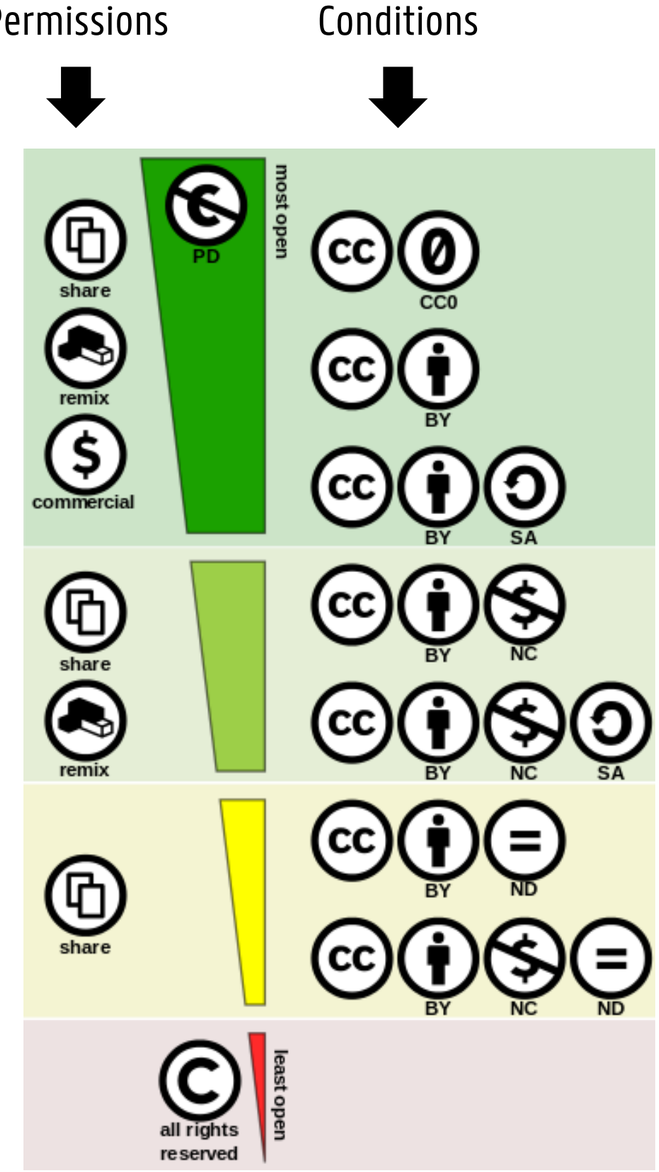
LICENSING: CREATIVE COMMONS LICENSES

- Public Domain Dedication (CC0 1.0)

  Waives copyright and related rights (e.g. databases).
- Attribution (CC-BY-4.0)

  Gives maximum freedom to reuse your work, provided they give appropriate credit.
- Attribution Share-Alike (CC-BY-SA-4.0)

  Requires redistribution of derivative works under this same license.



CC license selector

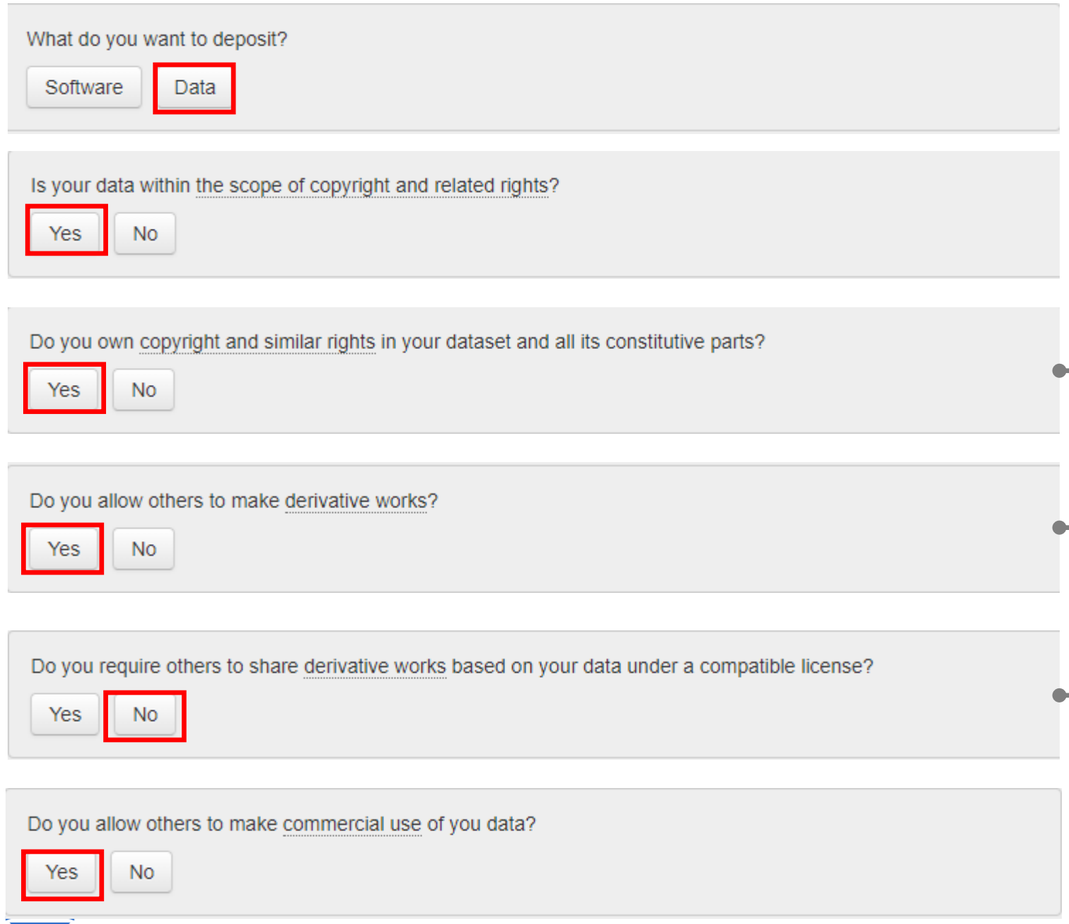
SHARING & LICENSING DERIVATIVE WORKS

- Whenever licensing a work or dataset that contains data already licensed, do not claim rights you do not hold.
- If the derived data was licensed with conditions:
  - Copyleft / SA → license the derived dataset in the same way.
  - NC → do not charge for the derived dataset.
- Tool: License selector http://ufal.github.io/public-license-selector/

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EXAMPLE LICENSE SELECTOR

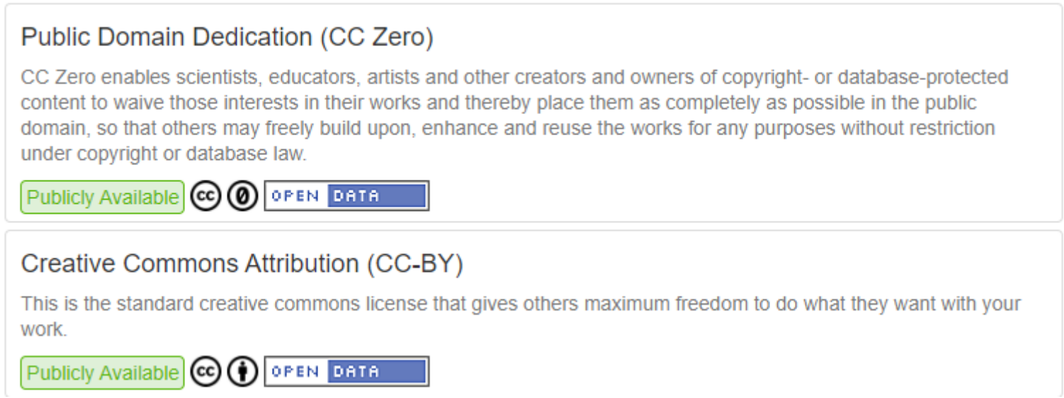
http://ufal.github.io/public-license-selector/



Implies you are not re-using data (your work does not include derivatives)

If no, would add the Non-Derivative clause

If yes, would add the copyleft or Share Alike clause



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EXAMPLE LICENSE SELECTOR

http://ufal.github.io/public-license-selector/



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TAKE HOME MESSAGES

Preserving data

- Select what to keep
- Usability
- Will need some effort
- Plan!

Sharing

- Consider sharing
- Use trustworthy repository
- Choose appropriate license
- Plan!

Thank you for attending!

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